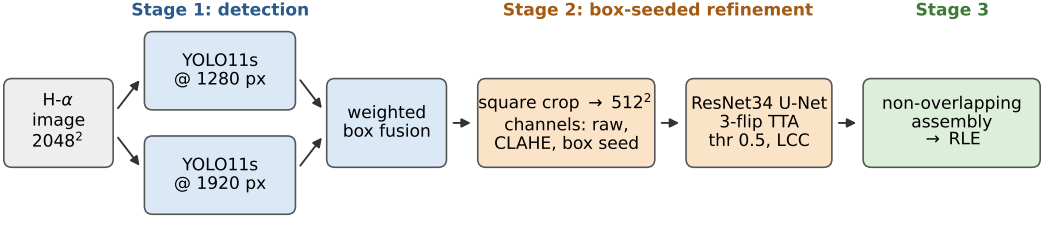
Stage 1: detection
Stage 2: box-seeded refinement
Stage 3
H-α image 2048²
YOLO11s @ 1280 px
YOLO11s @ 1920 px
weighted box fusion
square crop → 512² channels: raw, CLAHE, box seed
ResNet34 U-Net 3-flip TTA thr 0.5, LCC
non-overlapping assembly → RLE